

Problem G. Second Largest Query

Time Limit 2000 ms

Problem Statement

You are given a sequence $A = (A_1, A_2, \dots, A_N)$ of length N .

Process Q queries in the order they are given. Each query is of one of the following two types:

- Type 1: Given in the form $1 \ p \ x$. Change the value of A_p to x .
- Type 2: Given in the form $2 \ l \ r$. print the **number of occurrences** of the second largest value in $(A_l, A_{l+1}, \dots, A_r)$. More precisely, print the number of integers i satisfying $l \leq i \leq r$ such that there is exactly one distinct value greater than A_i among $A_l, A_{l+1}, \dots, A_r$.

Constraints

- $1 \leq N, Q \leq 2 \times 10^5$
- $1 \leq A_i \leq 10^9$
- For type-1 queries, $1 \leq p \leq N$.
- For type-1 queries, $1 \leq x \leq 10^9$.
- For type-2 queries, $1 \leq l \leq r \leq N$.
- There is at least one type-2 query.
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N Q
A1 A2 ... AN
query1
⋮
queryQ
```

Here, query _{i} is the i -th query and given in one of the following formats:

1 p x

2 l r

Output

Let q be the number of type-2 queries. Print q lines. The i -th line should contain the response to the i -th type-2 query.

Sample 1

Input	Output
5 4 3 3 1 4 5 2 1 3 2 5 5 1 3 3 2 2 4	1 0 2

Initially, $A = (3, 3, 1, 4, 5)$.

For the first query, the second largest value in $(3, 3, 1)$ is 1, which appears once in $3, 3, 1$, so print 1.

For the second query, there is no second largest value in (5) , so print 0.

The third query makes $A = (3, 3, 3, 4, 5)$.

For the fourth query, the second largest value in $(3, 3, 4)$, is 3, which appears twice in $3, 3, 4$, so print 2.

Sample 2

Input	Output
1 1 1000000000 2 1 1	0

Sample 3

Input	Output
8 9 2 4 4 3 9 1 1 2 1 5 4 2 7 7 2 2 6 1 4 4 2 2 5 2 2 7 1 1 1 1 8 1 2 1 8	0 1 0 2 4